

HAT-P-19b

R-1103

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241113 · created 2026-07-30T19:46:35+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460627.6559 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.0083
Transit depth (Rp/Rs)^2	3.17 +/- 0.29 [%]
Orbital Inclination (inc)	89.97 +/- 0.63
Airmass coefficient 1 (a1)	0.986 +/- 0.02
Airmass coefficient 2 (a2)	0.01 +/- 0.1
Scatter in the residuals of the lightcurve fit is	2.05 %
Best Comparison Star	#1 - [497, 298]
Optimal Method	PSF photometry
Transit Duration (day)	0.102 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.178 $\pm$ 0.0083 vs 0.1418 (deviation 4.19 σ)
Duration fit vs published	0.102 d vs 0.1196 d (deviation 5.62 σ)
Mid-time O-C	-12.8 min
Depth SNR	20.6
Residual scatter	2.05 %

Light curve

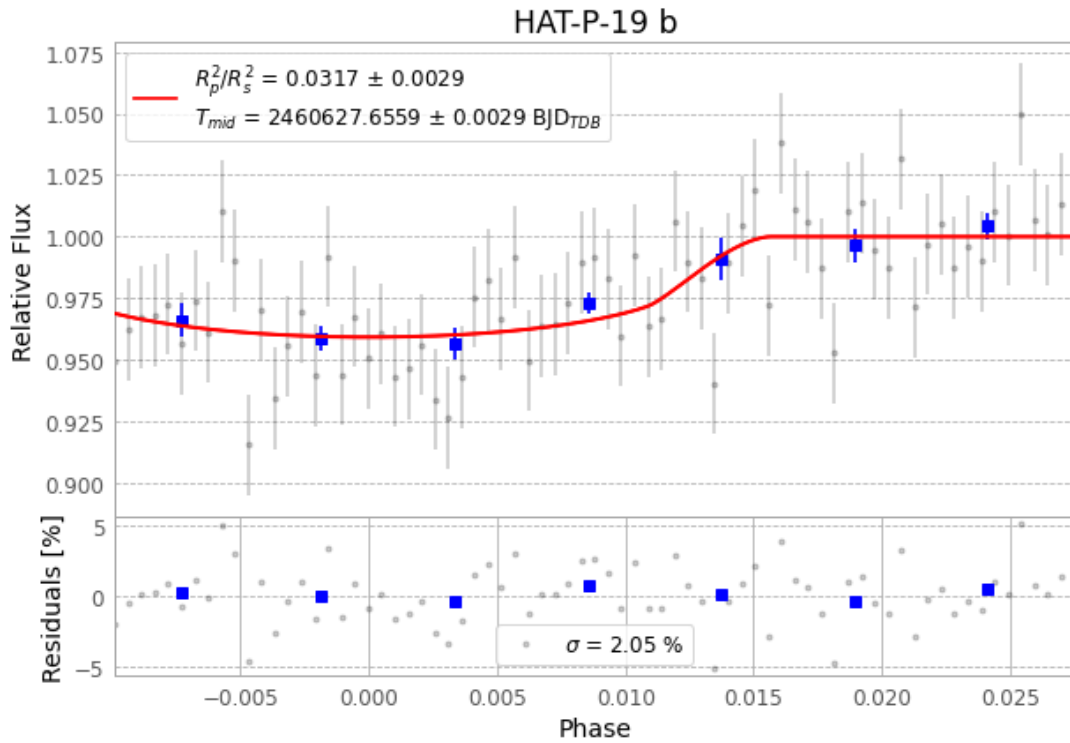


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-11-12.png (SHA-256 6530a0ea2b0aad19...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ccbbb4702d410aee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-19241113023917.FITS → HATP-19241113061515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241113.log (05b7aff6608a...), FinalLightCurve_HAT-P-19 b_2024-11-12.png (6530a0ea2b0a...), FinalLightCurve_HAT-P-19 b_2024-11-12.csv (53757860f899...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:46Z.